Serverless – Cloud Video Editor for Social Media

- **Theme- Industry, Innovation & Infrastructure (SDG 9)**
- **Category- Software**
- **Group No - 2**
- **Group Members- Prathamesh More, Krinjal Kocharmutha, Raj Patil, Harsh Kalsekar**
- **Guide- Prof.Vijaya Bharti**

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Introduction

- **Background**
- In the modern digital era, short-form video content such as reels, shorts, and clips has become one of the most dominant forms of communication and entertainment.
- Content creators often record long videos and later spend significant time manually editing them to extract engaging moments.
- This manual editing process is time-consuming, requires technical expertise, and often demands expensive software and high-end hardware.

- **Importance**
- Faster content creation for creators.
- Automation reduces workload.

- **Current Challenges**
- Identifying viral moments manually.
- Expensive editing software and GPUs.

Introduction

- **Motivation**
- Need for cost-effective automated editing.
- This project aligns with SDG 9 – Industry, Innovation, and Infrastructure by promoting innovative digital infrastructure and cost-effective technological solutions.
- It demonstrates how cloud-based services and AI tools can be used to create scalable, efficient, and affordable software systems.
- By using student free-tier accounts and serverless resources, the system encourages technology adoption among students and startups while reducing financial barriers.

- **Objectives**
- Automate highlight detection.
- Reduce editing time.
- Provide serverless scalable system.

Literature Survey:

Title	Conference / Journal Details	Key Points	Improvements Proposed
Cloud-Based Automatic Video Editing Using Keywords (2014)	ResearchGate	Keyword matching, cloud processing, automatic clip generation.	Relies only on keeywords, lacks AI sentiment analysis.
Automatic Video Summariation (2012)	IEEE ISM	Computer vision, frame extraction, scene detection.	GPU heavy, high computation cost.
AI-Based Multimedia Content Analysis (2020)	ScienceDirect	Machine learning & deep learning for video understanding.	Requires large datasets, expensive infrastructure, not cost-effective for students.
Deep Ranking Video Highlight Detection (2019)	ICCIDS	CNN + ranking models for highlight scoring.	Requires labeled training data, slow processing.
Intelligent Video Highlight Generation – HOMER (2021)	PMC	Emotion + video fusion for highlight detection.	Needs user emotion input, limited scalability.

Limitations of existing Systems

Traditional video editing tools require high-performance GPUs and paid software licenses, making them expensive and inaccessible to many users. Manual highlight detection is inefficient and subjective. AI-based editors often consume high computational resources, leading to slow processing and increased cost. These systems are also not optimized for beginners or student-level experimentation.

Problem statement and Objectives

- Manual video editing and highlight identification are inefficient, costly, and time-consuming.
  There is a need for an automated, cloud-based, cost-effective system that generates social-media-ready shorts instantly.

Objectives:-

- Detect high-sentiment moments automatically
- Reduce manual editing effort by 70–80%
- Eliminate GPU dependency
- Provide scalable cloud architecture
- Make system affordable for student

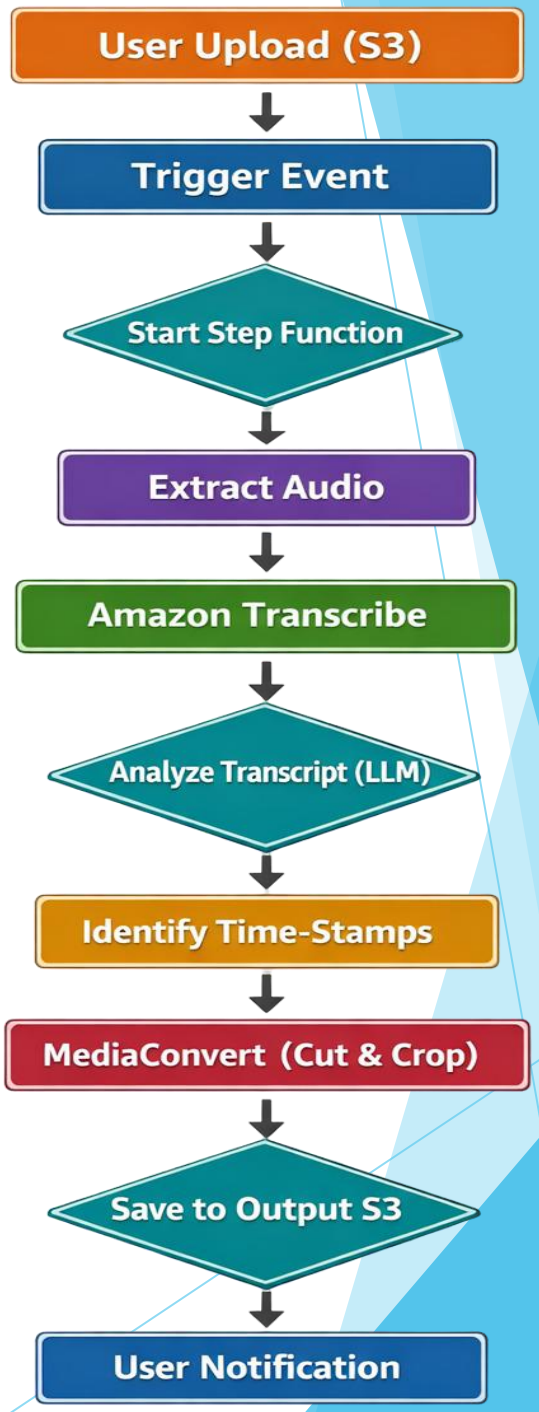
Proposed Methodology

Core Modules

- Upload Module (AWS S3)
- Processing Module (AWS Lambda)
- AI Module (Google Gemini API)
- Editing Module (FFmpeg)
- Output Module (S3 + Notification)

Innovation

- "Listen-to-Edit" approach using audio instead of heavy video AI
- Serverless architecture with zero idle cost
- Combines AWS + Generative AI



Experimental Setup

Input / Dataset

- User uploaded raw videos (.mp4 format)
- Sample videos of 5–30 minutes duration
- Audio extracted for transcription and sentiment analysis

Performance Evaluation Parameters

- Processing Time per Video
- Highlight Detection Accuracy
- Cost per Video Processing

Software / Technologies Used

- AWS Web Console
- AWS Lambda (Serverless Processing)
- AWS S3 Buckets (Storage)
- Vosk LLM (Audio Transcribe)
- FFmpeg (Audio Extraction & Video Cutting)
- Google Gemini API (Transcription & Sentiment Detection)

Hardware Setup

- Laptop / PC with Internet

Implementation

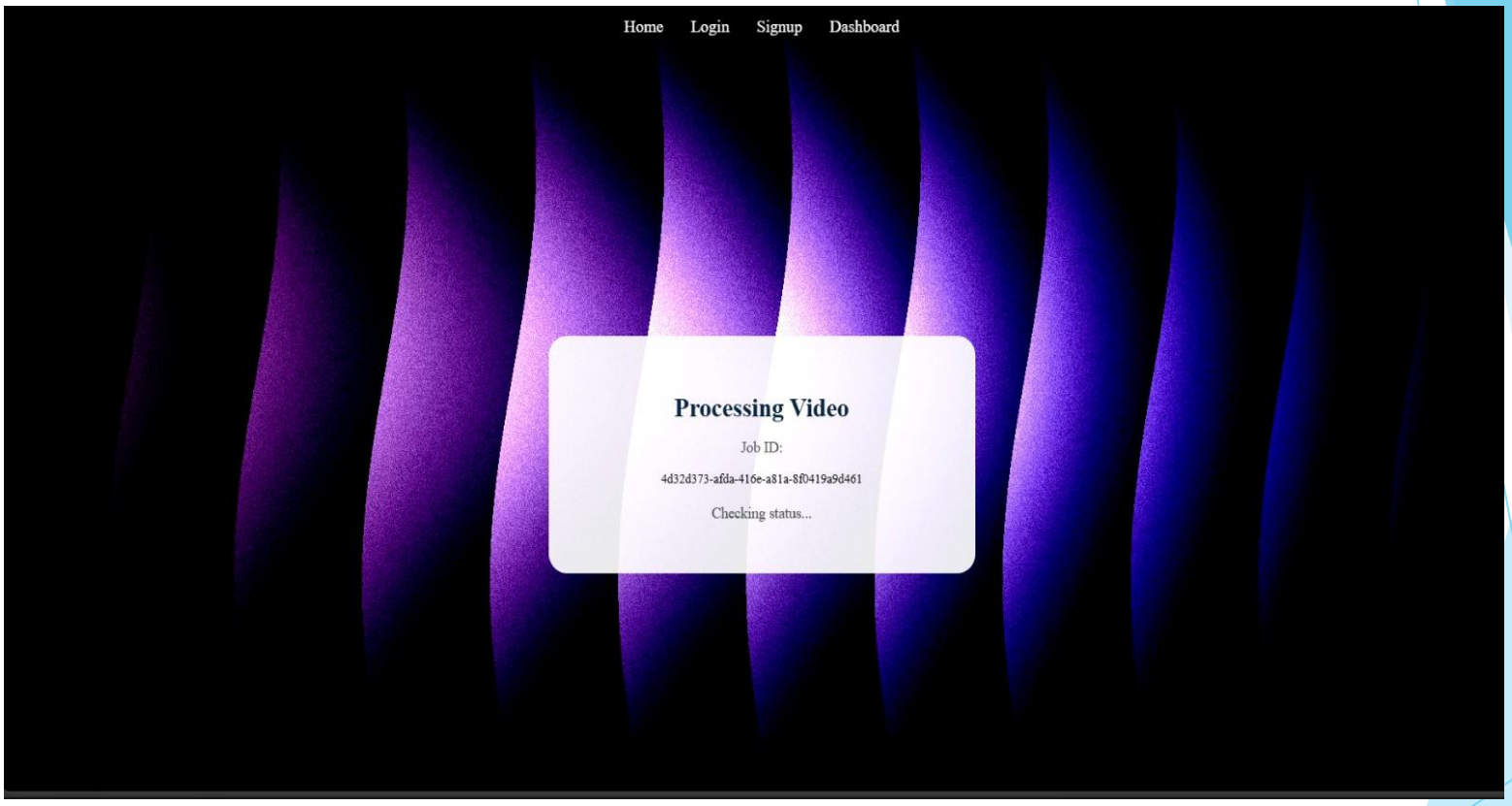


Fig1: DASHBOARD FOR UPLOADING VIDEOS

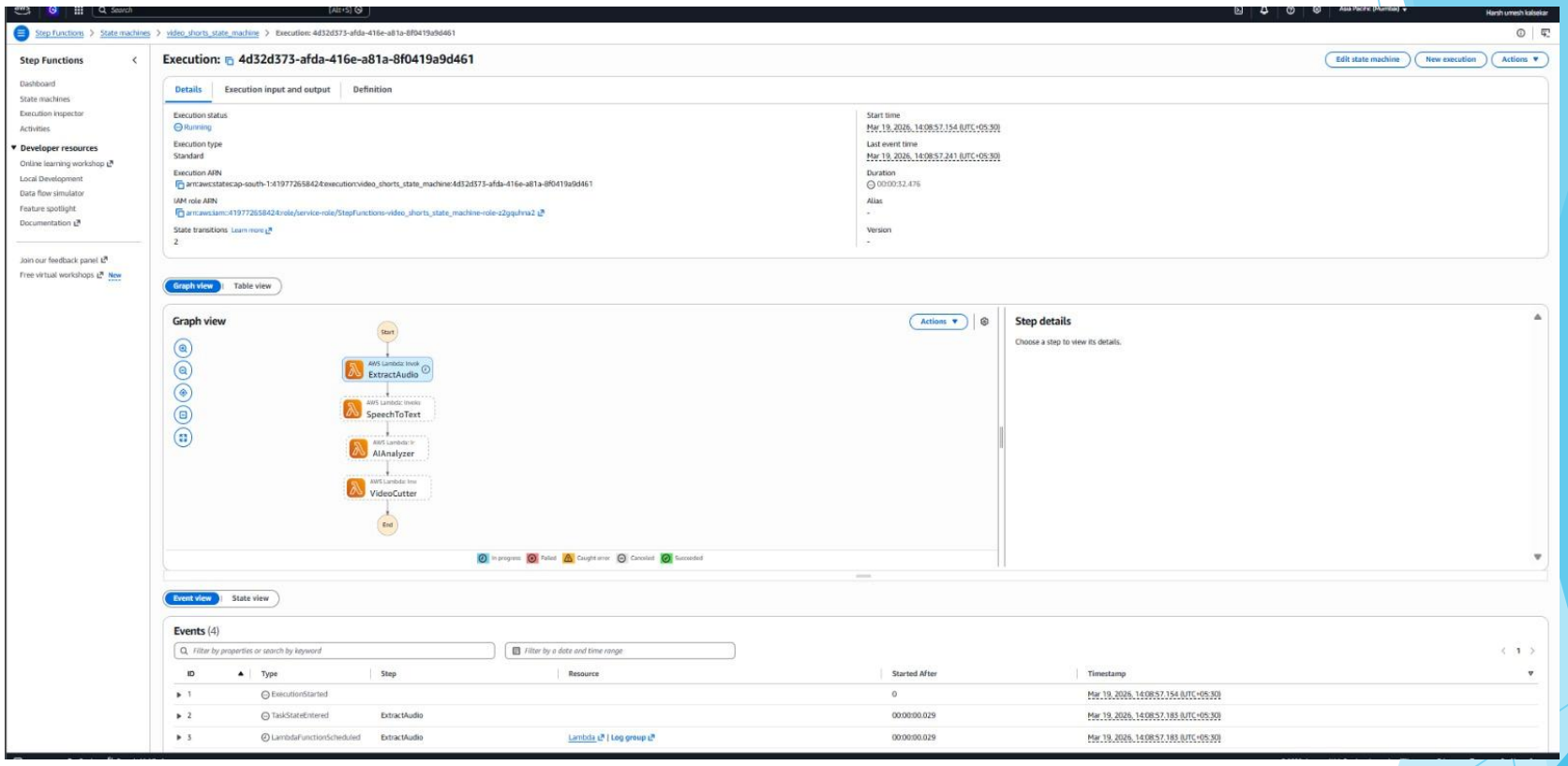


Fig2: AWS BACKEND PROCESS STEP MACHINE

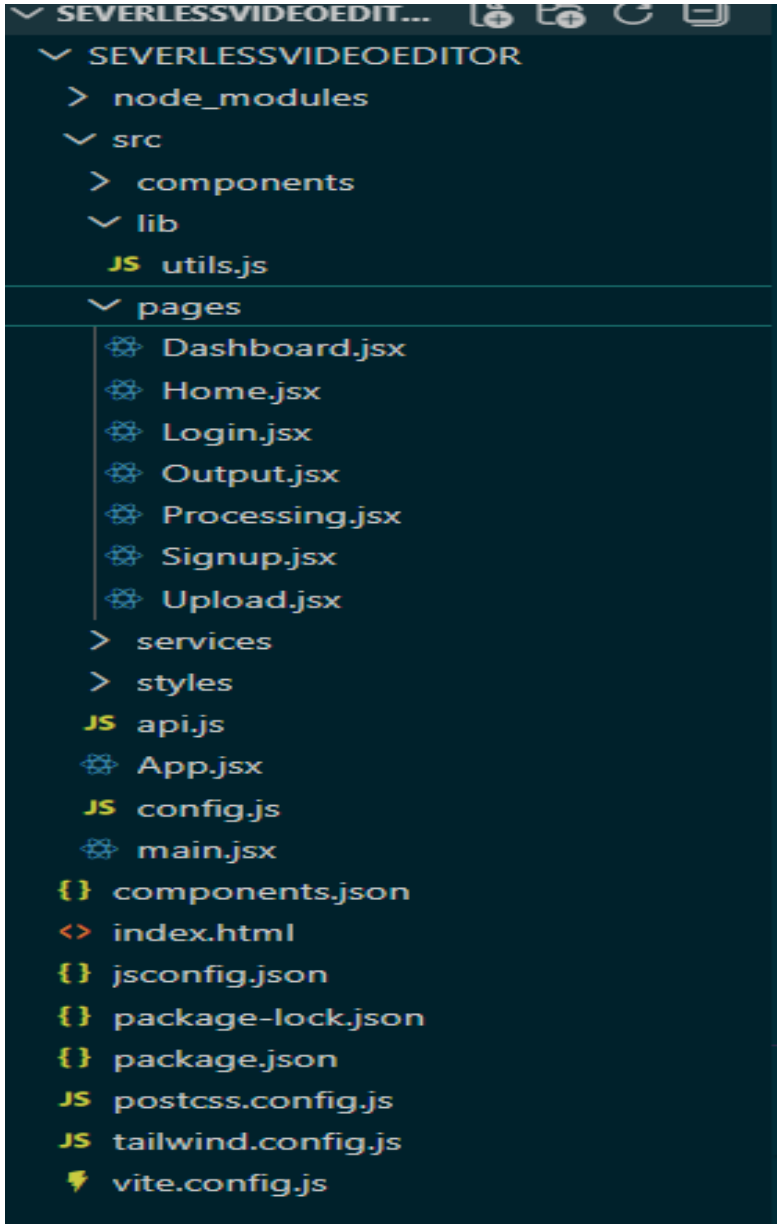


Fig 3:PROJECT FOLDER

Results & Analysis

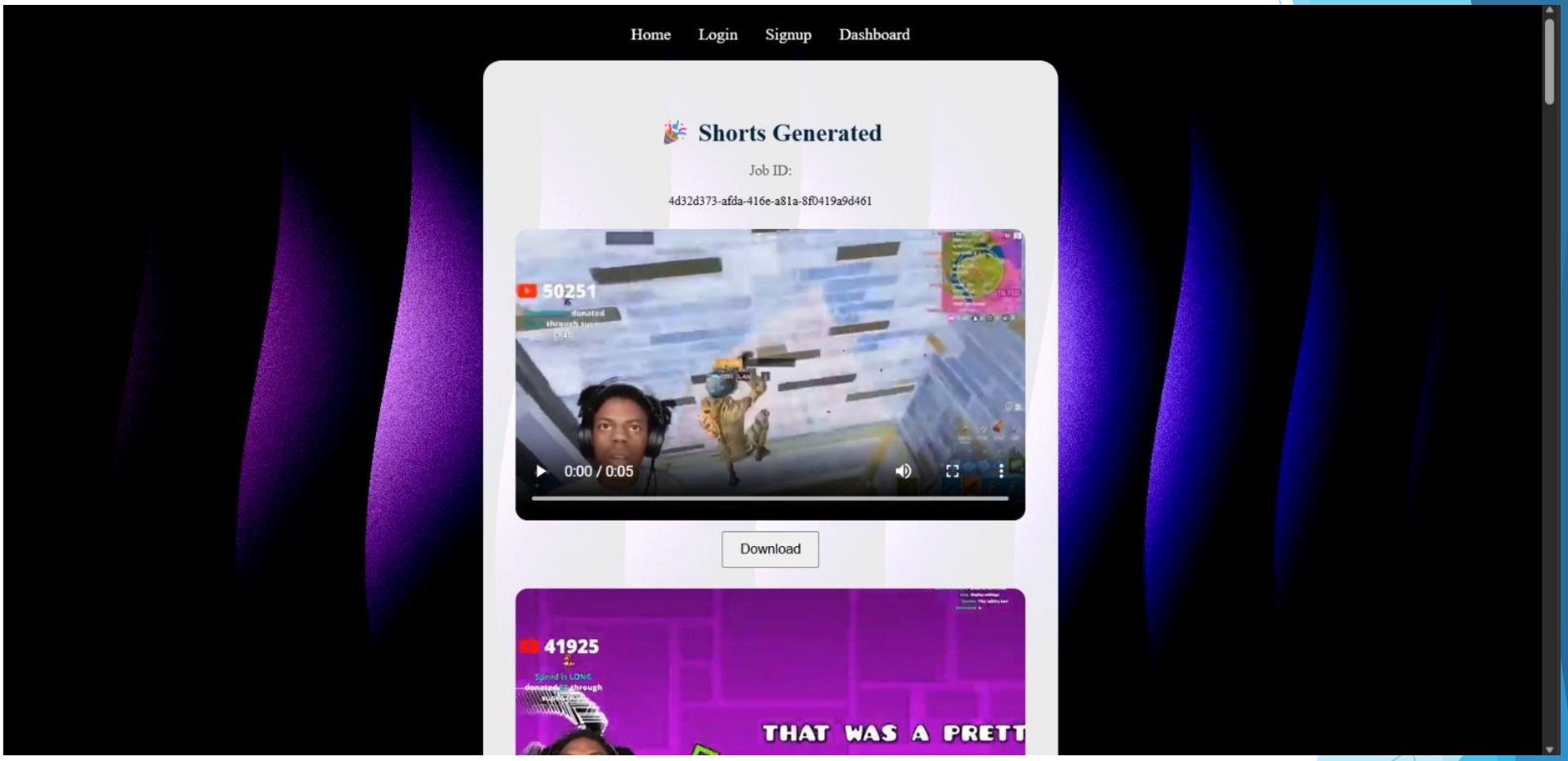


Fig 4:OUTPUT SCREEN

Conclusion and Future work

- The Serverless Cloud Video Editor successfully automates the process of extracting engaging short clips from long videos.
- It significantly reduces editing time and cost while eliminating the need for GPU hardware.
- The system demonstrates how serverless computing and generative AI can be combined to build scalable, efficient, and student-friendly software solutions.
- Overall, the project provides a practical and innovative approach to automated video editing.

Future Work

- Multi-language transcription
- Real-time editing
- Mobile app integration
- Advanced emotion detection
- Cloud auto-scaling improvements

References

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Thank You...!!